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HI-Captioner: End-to-end image captioning based on hierarchical multi-scale encoding and cross-modal interactive decoding

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ABSTRACT

The image captioning task aims to accurately describe images using natural language. It is widely applied in assisting visually impaired individuals, enhancing human–computer interaction, and image retrieval. However, existing end-to-end image captioning methods typically rely on single-scale feature extraction and use simplistic cross-modal interactions. This restricts the model's capacity to capture multi-level information and complex scenes, leading to captions that lack detailed expression and exhibit poor semantic coherence in context. In response, this paper proposes an innovative image captioning model HI-Captioner based on Swin Transformer to capture multi-scale features, hierarchical positional information and optimize inter-modal information interaction. HI-Captioner integrates Hierarchical Synergistic Attention Module (HSAM) and Cascading Cross-Modal Interaction Decoder (CCMID). The former combines a hierarchical multi-scale attention mechanism with a hierarchical positional encoding method during the encoding stage, enabling more effective capture of multi-scale features and multi-level positional information of images. This improves the model's ability to capture intricate nuances and comprehend global semantics in the process of generating captions. The latter adopts a novel cross-modal interaction approach to further enhance information exchange between image and text modalities during the decoding stage, significantly improving the model's efficiency in integrating and representing information from different modalities. Experimental results demonstrate that our proposed model yields substantial performance enhancements on the MSCOCO and Flickr datasets, particularly excelling in evaluation metrics such as BLEU4 and CIDEr. This demonstrates the great potential of the method in generating accurate and natural image captions. The code will be released at: https://github.com/Lwj-cap/HI_Captioner_main

1. Introduction

The image captioning task aims to generate text in natural language that accurately and thoroughly describes the content of an image. This task plays a crucial role in applications like helping visually impaired individuals comprehend image content [1], enhancing human–computer interaction experiences [2] and improving image retrieval efficiency [3,4]. With the swift advancements in computer vision and natural language processing technologies, especially with the introduction of Transformer models, substantial progress has been achieved in image captioning, particularly in terms of accuracy and coherence [5].

Although existing methods have achieved some success in image caption generation, they are still deficient in handling complex scenes and capturing detailed information [6]. Most existing end-to-end image captioning methods adopt an encoder–decoder architecture. In this approach, the encoder utilizes Convolutional Neural Networks (CNN) to extract visual features, while the decoder leverages Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM) networks, or Transformer-based architectures to produce captions [7–11]. The aforementioned methods exhibit deficiencies in encoding image features and capturing multi-level information, which hinder the model's effectiveness in extracting and integrating features across different levels, ultimately impacting its ability to generalize in subsequent generation tasks [12]. Additionally, existing positional encoding methods have limitations in representing the hierarchical relationships within images, impacting the accuracy and naturalness of the generated captions [13,14].

In the decoding process of image captioning, the decoder relies solely on previously generated words and image features when generating each word, while ignoring the potential influence of subsequent words on the current word [15]. This one-way flow of information could restrict the coherence and precision of the generated captions,

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making it difficult for the model to fully integrate the context in complex scenes [16]. Therefore, achieving more comprehensive information transmission and cross-modal interaction during the decoding process to enhance the coherence and naturalness of captions is a key objective in end-to-end image captioning tasks [17].

In this paper, we present an end-to-end captioning method based on Swin Transformer [18], which introduces a Hierarchical Multi-Scale Attention (HMSA) mechanism at the stage of processing the extracted image features. This mechanism uses multi-scale convolution at different layers to extract different granular features of the image. It also defines matching attention adjustment factors to adjust the attention weights at each layer, which enables the model to allocate attention more flexibly at different levels and effectively capture multi-level information of the image, enhancing the accuracy of image captioning and the richness of detail caption. Additionally, in order to complement the HMSA, a Hierarchical Positional Encoder (HPE) is proposed, which introduces Hierarchical Modulation Factors (HMF) to better capture positional information across different hierarchical sequences. HPE builds upon traditional positional encoding by incorporating a position scaling mechanism with hierarchical adaptive capabilities. This not only preserves the advantages of classical positional encoding in capturing relative positional relationships but also allows for flexible adjustment of the reliance on positional information at different network layers through the introduction of HMF. As a result, the model's feature representation capacity is significantly enhanced in sequence modeling tasks, especially in balancing the global semantic integration and local detail capturing. HMSA and HPE together constitute the HSAM. In order to enhance the interaction between the two modalities of image and text in the captioning stage, we propose a cascaded cross-modal interactive decoding (CCMID). CCMID can accurately capture and fuse the complex details and high-dimensional semantic relationships between modalities, which significantly improves the efficiency of information integration and representation in cross-modal tasks, and achieves more comprehensive information interaction.

In this study, we applied the HSAM and the Cascading CCMID to an end-to-end image captioning model, introducing HI-Captioner. We extensively evaluated our model on the MSCOCO benchmark dataset, and a series of experiments demonstrated its effectiveness. The key contributions of this work can be summarized as follows:

- The HI-Captioner model introduced in this paper is capable of capturing both multi-scale features and hierarchical positional information, overcoming the limitations of existing methods in handling complex scenes and multi-level information representation. It enhances the model's precision and flexibility in capturing relative positional information and global contextual information within sequences, enabling deeper interaction between the image and text modalities.
- In the encoding stage, HSAM and HPE work together in capturing local details and global semantics, enabling the model to generate image captions that preserve the richness of details while ensuring global coherence and naturalness in generating image captions.
- In the captioning stage, the CCMID employs a cascading strategy to integrate Self-Attention and Cross-Attention mechanisms. This enables cross-modal information interaction, further strengthening the model's capability to capture both fine-grained details and global semantic information in multimodal tasks.
- The performance of HI-Captioner was validated on the MSCOCO benchmark dataset, applying the multi-level information fusion module and the bidirectional attention decoding mechanism to the Swin Transformer-based image captioning model. The model's robustness was further tested on the Flickr8k and Flickr30k datasets, achieving optimal results in BLEU4 and CIDEr evaluation metrics across all these datasets.

2. Related work

2.1. Image caption models

Image captioning models are generally divided into two categories: two-stage and end-to-end models. Two-stage models separate the task into feature extraction and caption generation [19–21]. Commonly, CNNs or Faster R-CNN [22] extract features, which are then fed into models like LSTM or Transformer with attention mechanisms. Several advanced approaches, such as the Dual-Level Collaborative Transformer [23], X-linear attention network [24], and Semantic-Conditional Diffusion Networks [25], attempt to improve semantic modeling. However, two-stage models cannot achieve end-to-end optimization and may lose fine-grained details, making them less efficient and expressive. To address challenges such as limited semantic understanding and poor fluency, researchers have explored various strategies. Some incorporate structured knowledge (e.g., scene graphs, syntactic trees) to enhance cross-modal alignment and cross-lingual fluency [26], while others design dynamic networks like DTNet with adaptive routing [27]. Additional efforts include targeting hard examples via focal mechanisms and adaptive reward functions like ECIDEr [28], as well as improving attention mechanisms through spatial-channel semantic modeling as in SDATR [29].

End-to-end image captioning models unify feature extraction and caption generation within a single framework, enabling direct image-to-caption mapping and joint optimization. This design improves caption precision by capturing fine details and reduces training complexity and resource requirements [17,30,31]. Early models used CNNs for visual encoding and LSTMs for decoding. With advances in CNNs (e.g., ResNet, Inception, VGG) and the emergence of vision Transformers [32,33], feature extraction has become more effective. However, CNNs struggle with global semantics, while Transformers may miss fine-grained details. On the decoding side, efforts include refining RNN or LSTM decoders [9,11], adopting Transformer decoders, and integrating self-cross-attention. However, RNNs face gradient vanishing in long sequences, while Transformers, while adept at global modeling, may reduce the accuracy of details in complex scenes. Although significant progress has been made, there are still limitations in semantic and detail modeling. The end-to-end model with unified training and better granularity is a promising direction. Therefore, the focus of this work is to develop a robust end-to-end image captioning model to improve multi-scale semantic representation and cross modal fusion.

2.2. Attention and location encoding methods

Attention mechanisms have become a core technique for enhancing performance in end-to-end image captioning [34–36]. With the adoption of Transformer architectures, attention mechanisms have further improved the modeling of global semantics and fine-grained details. However, the self-attention mechanism in Transformers struggles to capture both local and global information due to its fixed receptive field. To address this, various improved attention mechanisms have been proposed to process visual features more effectively across different scales [37], enhancing caption detail and accuracy. For instance, Zhuang et al. introduced multi-scale Transformer decoding with a multi-scale language decoder for denser captions [38], while Jia et al. proposed a guided token attention network that reduces computational load while preserving global semantics [39]. These methods enable models to better attend to multi-scale features, especially in complex scenes with multiple objects, leading to more precise and semantically rich captions.

Positional Encoding is an essential method in Transformer models used to capture positional information in sequence data. In image captioning tasks, Positional encoding assists the model in comprehending spatial relationships between features in an image. By combining improved attention mechanisms with positional encoding, the

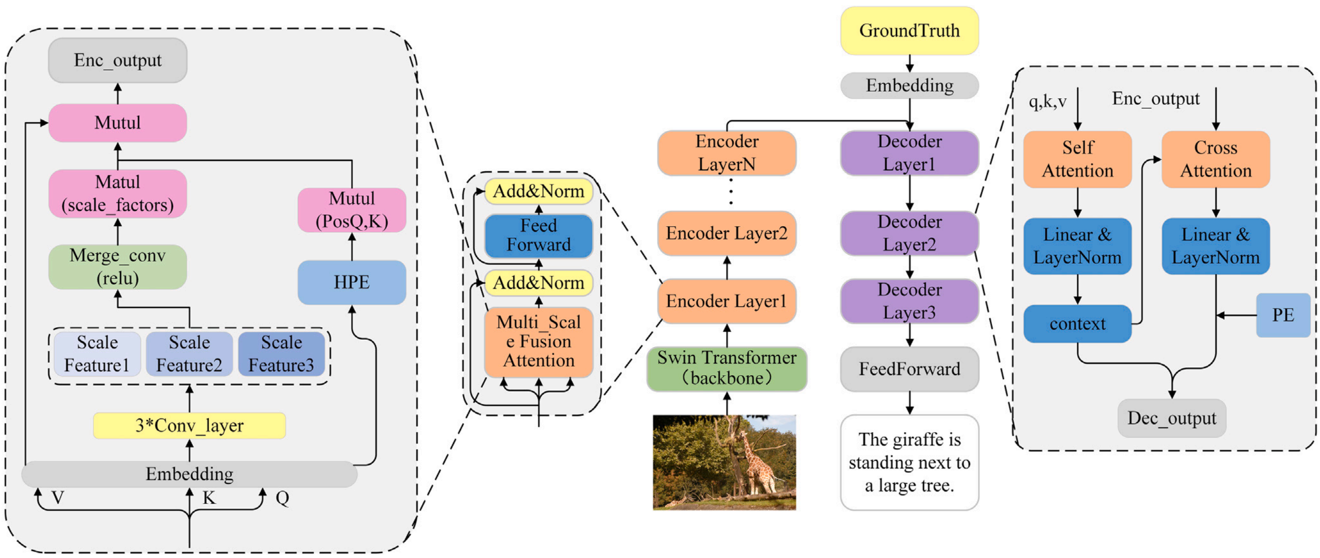

Fig. 1. Schematic overview of the proposed HI-Captioner model. The architecture consists of a Swin Transformer-based visual encoder with multi-scale fusion attention, and a Transformer decoder for caption generation. The encoder extracts multi-level features and integrates them via the Mutul module. In the decoder, self-attention is applied over previous tokens, and cross-attention is applied over encoded image features. Here, Q, K, V denote query, key, and value vectors of the text input; PE and HPE represent standard and hierarchical positional encodings, respectively.

model exhibits stronger feature representation capabilities and achieves higher generation quality [40]. For example, Wang et al. proposed the geometry attention Transformer model, which integrates a geometric gating mechanism with position-aware LSTM during both the encoding and decoding stages, improving the quality of generated captions [41]. Liang et al. modified the decoding approach to a masked recovery method and introduced a multi-head attention module with relative positional encoding, enabling the model to better capture the relative positional relationships between words, thereby enhancing image captioning performance [42]. However, such methods fail to fully utilize and differentiate hierarchical features when applying positional encoding to multi-scale features, resulting in relatively simplistic positional information capture.

To address the limitations in capturing hierarchical semantic details and effectively modeling positional information across multiple scales, we propose a Hierarchical Multi-scale Attention Mechanism and Hierarchical Positional Encoding.

2.3. Cross-modal interaction

Early multimodal tasks typically considered forward contextual information only within a single modality, which prevented the model from fully leveraging the interconnections and information fusion between multiple modalities. Traditional cross-modal interaction methods also performed inadequately when handling complex scenes, struggling to capture the finer details of input sequences and generate coherent outputs. In recent years, Cross-modal interaction methods have made significant advancement in multimodal tasks [43]. In image captioning tasks, cross-modal interactive decoding enables better utilization of contextual information from both image and text modalities, improving the semantic coherence and richness of details in the generated captions [44]. For example, the multimodal Transformer model [45] uses directional pairwise cross-modal attention to adapt information from one modality to another, achieving optimal cross-modal information capture. Moreover, this approach has been widely applied to other multimodal tasks such as visual question answering [46] and video captioning [47], further demonstrating its effectiveness in cross-modal information fusion. However, the existing cross-modal interaction methods do not capture local details and global semantics in a balanced way, and the information transfer between modalities is insufficient when dealing with long-distance dependencies and complex relationships, which leads to a lack of semantic coherence in the generated text. This issue is especially pronounced in long-sequence generation, where breaks or repetitions are more likely to occur.

Therefore, we design a new decoding mechanism to alleviate the limitations of current cross-modal interaction methods, aiming to achieve higher-quality caption generation. This mechanism dynamically fuses textual and visual features during decoding, and its hierarchical fusion enables gradual and coherent integration of visual and linguistic modalities, particularly benefiting long-sequence generation in complex scenes.

3. Methods

3.1. Overview

This paper adopts an end-to-end encoder–decoder architecture based on Swin Transformer for image captioning. The encoder first extracts multi-scale visual features, which are enhanced by the Hierarchical Multi-Scale Attention Mechanism (HSAM) to capture both fine-grained details and global semantics. Meanwhile, the Hierarchical Positional Encoder (HPE) injects adaptive positional information across scales to improve spatial understanding. These enriched features are then fed into the Transformer decoder. During decoding, the Cascaded Cross-modal Interaction Decoder (CCMID) performs gated cross-attention between image features and partial text context at each layer, enabling progressive and coherent visual-text alignment. This design ensures accurate and detailed caption generation grounded in both local and global visual cues. The enhanced Transformer architecture includes 9 encoder layers and 3 decoder layers, as shown in Fig. 1. Here Q, K, and V are the results derived by applying parameter matrices W_Q, W_K, and W_V to project the input sequence X. The hierarchical positional encoding (HPE) is applied to the multi-scale features to retain spatial structure before cross-modal alignment.

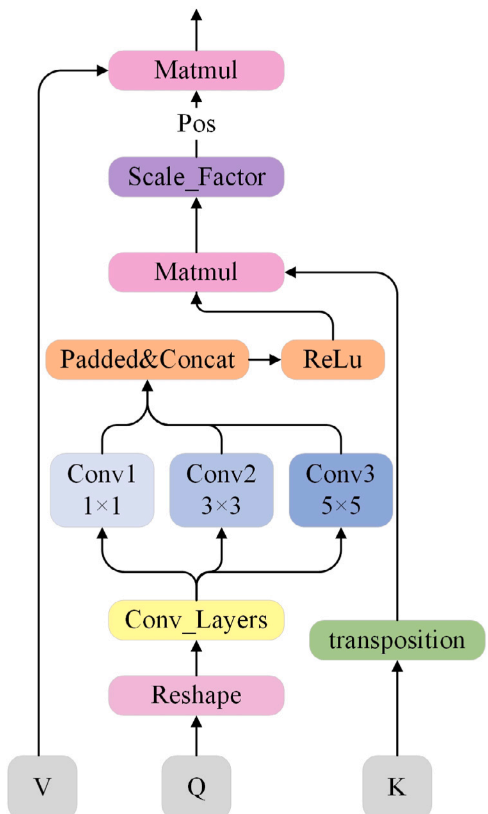

Fig. 2. Architecture of the proposed Hierarchical Multi-Scale Attention (HMSA) module. The module takes query (Q), key (K), and value(V) as input. Q is reshaped and passed through multi-scale convolution layers (1 × 1, 3 × 3, 5 × 5) to capture local context at different receptive fields. The outputs are concatenated and activated via ReLU, followed by a dot-product attention mechanism scaled by a learned scale factor. K is transposed before attention computation. The positional encoding is applied prior to the final matrix multiplication.

3.2. Hierarchical synergistic attention module

The central element of this model presented in our paper is the Hierarchical Synergistic Attention Module (HSAM), designed to enhance the ability to capture and represent multi-level information in images. HSAM is composed of two components: the Hierarchical Multi-Scale Attention (HMSA) and the Hierarchical Positional Encoder (HPE). The multi-scale convolution and positional encoding in the shallow layers emphasize the fine-grained processing of local information, while the global convolution and enhanced positional encoding in the deeper layers help the model better integrate global information. The working principles of these two modules are detailed below.

3.2.1. Hierarchical multi-scale attention

The Hierarchical Multi-Scale Attention (HMSA) module extracts multi-granularity features from images through multi-scale convolution. Each convolutional layer has different kernel sizes, allowing the model to capture features ranging from fine details to overall structures at varying granularities. The attention modulation factors adjust the features in attention computation according to the hierarchical level, enabling the model to flexibly fuse features and capture contextual information at different levels. In the shallow layers, the model focuses on local details, and the modulation factor is larger to increase the influence of local features on the final outcome. In contrast, the deeper layers focus on global information and long-distance dependencies, where the modulation factor is smaller to prevent global features from excessively influencing the outcome. The structure of HMSA is illustrated in Fig. 2.

$$\text{scale\_factor} = \alpha^{i} \cdot \sqrt{d_k} \tag{1}$$

The input features are initially reshaped before being processed through the multi-scale feature extraction layer to obtain the multi-scale features M. q represents the input features after dimensional transformation, and n represents the quantity of multi-scale layers. The Eq. (1) is as follows:

$$M[i] = \text{Conv}_i(q) \quad i \in (1, n) \tag{2}$$

After obtaining the multi-scale features *M[i]*, the ReLU activation function is utilized to derive the fused feature representation q_{layer}, and an attention adjustment factor is introduced to modulate the attention computation. Pos represents the contribution of the positional encoding. The attention score is computed using the Eq. (2):

$$\text{scores} = \frac{\text{layer} \cdot k^{T}}{\text{scale\_factor}} + \text{Pos} \tag{3}$$

The *scale_factor* is calculated based on different levels using an exponential decay method. α is a decay coefficient less than 1, and $\sqrt{d_k}$ is used as the initial adjustment factor. With the growing number of layers, the adjustment factor gradually decreases. The calculation formula is Eq. (3):

3.2.2. Hierarchical positional encoder

The proposed Hierarchical Positional Encoder (HPE) generates positional encoding by combining sine and cosine functions. To correspond with the local and global features extracted by HMSA, a position modulation factor (PMF) is introduced. In the positional encoding, PMF increases with the depth of the layers, allowing the shallow layers to emphasize the fine-grained expression of local information, while gradually increasing the reliance on global features in deeper layers. This helps the model better integrate global semantics and long-range dependencies at higher levels, thereby enhancing its understanding of positional information in the sequence. The specific procedure for positional encoding is outlined as follows:

Firstly, the query Q and key K are encoded using Hierarchical Position by the Eq. (4):

$$\begin{aligned} \text{Pos}_Q &= HPE(Q) \\ \text{Pos}_K &= HPE(K) \end{aligned} \tag{4}$$

The similarity between feature pixels is then captured using matrix multiplication to reflect the spatial positional relationship of each feature pixel within a header by the Eq. (5):

$$\text{Pos}_{QK} = \text{Pos}_Q * \text{Pos}_K{}^{T} \tag{5}$$

During the forward propagation process, the positional encoding is scaled and added to the input features according to the conditioning factor, which is calculated as follows for M_{Factor} by the Eq. (6):

$$M_{\text{Factor}} = \frac{\text{layer\_idx} + 1}{\text{total\_layers}} \tag{6}$$

Pos represents the final positional encoding result, which is fused during the attention computation process:

$$\text{Pos} = \text{Pos}_{Qk} \times M_{\text{Factor}} \tag{7}$$

In order to represent the HPE method more clearly, Algorithm 1 illustrates its basic computational procedure.

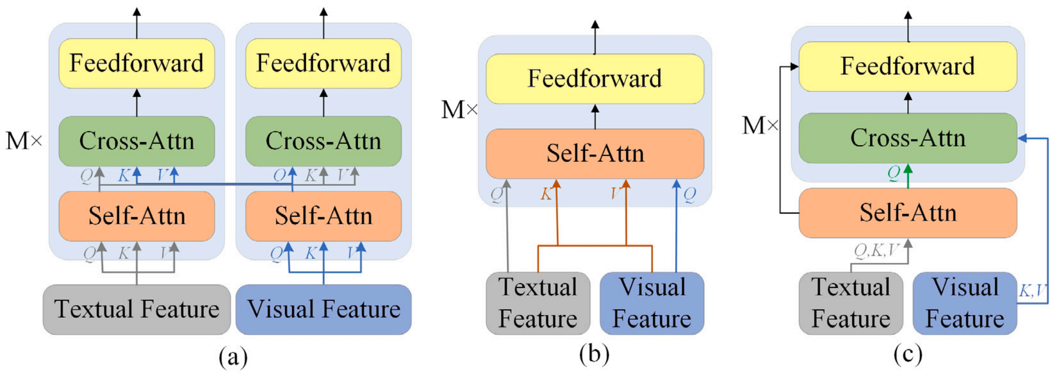

Fig. 3. Illustration of different intermodal interaction decoders. (a) Co-attention decoder: visual and textual features are processed in parallel via separate self-attention and cross-attention branches. (b) Merged attention decoder: visual and textual features are concatenated and processed jointly using a shared self-attention mechanism. (c) The proposed CCMI decoder: textual features undergo self-attention, and then interact with visual features through a unified cross-attention mechanism.

Algorithm 1: Hierarchical Positional Encoder (HPE)

Input: Feature matrices of different scales Q, K
Output: Pos
Input the multi-scale feature matrices Q and K obtained from Eq.(1) into Eq.(4) for calculation.
Assuming the input sequence length is T, the embedding dimension is d_model, and the position encoding is calculated using the following formula.

$$PE_{(pos,2i)} = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$$

$$PE_{(pos,2i+1)} = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{model}}}}\right)$$

The Position Modulation Factor gradually increases according to Eq.(6).
Compute the position score layer by layer by Modulation Factor.
posq = self.HPE (Q).
posk = self.HPE (K).
for p1, p2 **in** enumerate (posq, posk):
    Pos = torch.matmul (p1, p2.transpose (−1, −2)).
return Pos.
Fuse the Pos with the attention result using Eq.(2).

3.2.3. Cascade cross-modal interaction decoder

This paper adopts a cascading cross-modal interaction decoding (CCMID) approach for caption generation. This architecture utilizes self-attention and cross-attention mechanisms to sequentially process the input text sequence and image features, fully capturing contextual information and enabling efficient information exchange between modalities. To further demonstrate the effectiveness of CCMID, we compared it with two other common interaction methods, with the results presented in Section 4.3. Fig. 3 illustrates the attention interaction mechanisms of collaborative attention [48], merged attention [49], and the proposed CCMI attention interaction. In collaborative attention, the text and visual features are fed into separate transformer blocks, where self-attention and cross-attention are applied independently for cross-modal interaction. Merged attention, on the other hand, concatenates the text and visual features first, then feeds them into a single transformer block for interaction. Our proposed CCMID decoder first processes the text input through a self-attention layer, and the output is then combined with the image encoder's output as input to the cross-attention module. This method reduces computational costs while achieving more efficient and comprehensive multimodal information interaction. The specific computational process is detailed below.

Both self-attention and cross-attention are realized by the multiple attention mechanism as shown in Eq. (8):

$$\text{Attention (Q,K,V)=SoftMax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \tag{8}$$

The input of self-attention is shown in Eq. (9), which is implemented to compute the attention on the current input sequence of the decoder and capture the contextual relationships within the sequence.

$$\begin{aligned} Q &= W_Q \cdot d_i \\ K &= W_K \cdot d_i \\ V &= W_V \cdot d_i \end{aligned} \tag{9}$$

The input of cross attention is shown in Eq. (10), using the output d_{eo} of the encoder functions as K and V, and the self-attention output of the decoder from the preceding stage is used as Q to compute the cross attention, thus enabling inter-modal information interaction.

$$\begin{aligned} Q' &= W_Q \cdot d_{so} \\ K' &= W_K \cdot d_{eo} \\ V' &= W_V \cdot d_{eo} \end{aligned} \tag{10}$$

The calculation of d_{so} is shown in Eq. (11). The decoder firstly receives the candidate subtitle sequence for text embedding and position encoding operations, and learns it through the self-attention mechanism, where d_i is the image subtitle sequence, d_l is the length of the input subtitle, and *target_emb ()* is the target vocabulary embedding function, which converts the vocabulary ID into a dense vector representation. *pos_emb ()* is the position encoding function that allows the model to identify various positions within the input sequence.

$$d_{so} = \text{target\_emb}(d_i) + \text{pos\_emb}(d_l) \tag{11}$$

Self-attention achieves understanding the own structure of the target sequence, and cross-attention combines the target sequence with the global context of the input sequence. The final output from the decoder is passed through a linear layer to create a probability distribution over the target vocabulary, resulting in the generation of the output sequence. The output of cross attention and the output of self attention are fused by multiple residual joins and layer normalization, as shown in the feedforward neural network Eq. (12), where b1 and b2 are the biases,W1 and W2 are the two sets of learned weight matrices.

$$FFN(x) = ReLU(xW_1 + b_1)W_2 + b_2 \tag{12}$$

3.3. Objective function

The model's performance is enhanced by the Cross-Entropy Loss [50], which assesses the difference between the predicted probability distribution and the true distribution of target labels. Through the process of minimizing this loss, the model is able to iteratively refine its parameters, which helps it produce captions that are both contextually relevant and semantically coherent based on the input image. For each time step t, the decoder yields an output in the form of a probability distribution, and the target output is the actual word y_t (the index of the target word). The cross-entropy loss can be defined as Eq. (13), where V is the size of the vocabulary. $\hat{y}_t[i]$ is the probability of the ith word in the predicted probability distribution, and $1_{[y_t=i]}$ is an indicator function that takes the value 1 if the target word is the ith word in the vocabulary, and 0 otherwise.

$$L_{CE}(\hat{y}_t, y_t) = -\sum_{i=1}^{V} 1_{[y_t=i]} \log(\hat{y}_t[i]) \tag{13}$$

For the entire sequence length T, the total loss of the decoder is the average of the losses across all time steps:

$$L_{total} = \frac{1}{T}\sum_{t=1}^{T} L_{CE}(\hat{y}_t, y_t) \tag{14}$$

To train the entire model, we take the average of the losses across all time steps for all samples in the batch, resulting in the overall loss function (15) for the model. Here B is the batch size, T_b is the sequence length of the b-th sample in the batch, and V is the size of the vocabulary. $\hat{y}_t{}^b[i]$ is the probability of the ith word for the b-th sample at time step t predicted by the decoder, and $\hat{y}_t{}^b[i]$ is the actual word index of the b-th sample at time step t. The final objective function is as follows:

$$L = \frac{1}{B}\sum_{b=1}^{B}\frac{1}{T_b}\sum_{t=1}^{T_b} -\sum_{i=1}^{V} 1_{[y_t^b=i]}\log(\hat{y}_t^b[i]) \tag{15}$$

4. Experiments

4.1. Data and evaluation indicators

Due to the advantages of the MSCOCO dataset in terms of the accuracy and diversity of image-text alignment, it is commonly utilized in the area of image captioning [51]. This dataset is available to the public, and to ensure consistent experimental results, we utilized the MSCOCO dataset following the splitting approach outlined by Karpathy [52], which separates the dataset into 113,287 images for training, 5000 for validation, and 5000 for testing, with each image linked to 5 corresponding captions.

In terms of evaluation metrics, we employed several commonly used benchmark metrics, including BLEU [53], METEOR [54], ROUGE-L [55], CIDEr [56], and SPICE [57] to assess the model's performance. The BLEU metric primarily assesses the n-gram correspondence between the generated image captions and the reference captions. METEOR, building on n-gram matching, further considers syntactic structure, word order, and morphological variations, providing a more nuanced evaluation. ROUGE-L emphasizes the longest common subsequence between the generated text and the reference text to assess the level of overlap and coherence within the sentences. CIDEr assesses the quality of the captions by calculating the TF-IDF weighted n-gram similarity between the generated caption and multiple reference captions. SPICE emphasizes semantic similarity by comparing the semantic tuples (objects, relationships, and attributes) of the generated and reference captions. This study refers to these metrics as B-4, M, R-L, C, and S in that order.

4.2. Experimental settings

Software and Hardware Setup: We utilized an NVIDIA RTX 3090 GPU to speed up the training process, alongside CUDA version 11.0. The programming was done in Python, and the implementation was carried out using the PyTorch framework, specifically Torch version 1.70.

Table 1
Comparison experiments for each component,√ indicates the introduction of this improvement into the model.

Model	HMSA	HPE	CCMID	B-4	M	R-L	C	S
Baseline				37.2	28.0	56.9	118.3	21.2
Ours	√			38.1	28.2	57.2	120.4	21.5
Ours	√	√		38.9	28.5	57.5	121.2	21.6
Ours	√	√	√	**39.4**	**28.7**	**58.1**	**122.3**	**22.3**

Table 2
Experimental results at different layer scales.

Scale	B-4	M	R-L	C	S	Time/image
2	37.3	28.1	57.0	118.4	21.2	**0.232 s**
3	38.1	**28.2**	**57.2**	**120.4**	21.5	0.258 s
4	**38.2**	**28.2**	57.1	120.2	**21.6**	0.342 s
5	37.4	28.0	57.1	119.3	21.3	0.384 s

Model Construction: We employed a pre-trained Swin Transformer as the foundational network. To conserve computational resources, pre-trained Swin Transformer weights were loaded during the training phase [31]. Following the settings in existing studies [17], the captioning network was configured with 9 Transformer layers in the encoder and 3 layers in the decoder.

Training Details: The main training dataset consisted of images and captions from the MS COCO dataset. The input image resolution was configured to 384 × 384. The standard Transformer was configured to utilize 8 attention heads. The AdamW optimizer [58] was used for optimization. During training, the batch size was set to 40, the learning rate to 1e-6, and the model was trained for 3 epochs. Additionally, to prevent overfitting, dropout was applied with a rate of 0.1. During sentence generation, a beam search algorithm with a beam width of 5 was utilized to identify the optimal word sequence [7,59]. The cross-entropy loss function was applied in all experiments conducted. No learning rate scheduler was applied during training, as the learning rate was fixed throughout.

4.3. Ablation experiments

4.3.1. Evaluation of the performance of the proposed module

In the HI-Captioner model, we proposed three different improvements. To assess the individual contributions of each component to the model's performance, we provide performance results obtained by sequentially incorporating each element into the baseline model, as shown in Table 1. The information presented in the table shows that the hierarchical multi-scale attention mechanism effectively improves the BLEU4, ROUGE-L, CIDEr metrics, resulting in an improvement of 0.9% and 2.1%, respectively. This demonstrates that our multi-scale attention mechanism plays a significant role in extracting and integrating visual representations, resulting in generated captions that are more aligned with human annotations. Additionally, the HPE module boosts the BLEU4 metric by 0.8%, while the CCMID module increases the CIDEr metric by 1.1%. When compared to the baseline model, our model improves BLEU4 by 2.2% and CIDEr by 4.0%.

4.3.2. Analysis of HMSA modules

The multi-level feature extraction method discussed in this study facilitates the extraction of information and fusion at different scales, allowing for a deeper insight into image content. Through experimental validation (Table 2), considering both the training accuracy and time of the overall model, we selected 3 as the optimal number of multi-scale layers.

Additionally, we assessed the caption generation effectiveness of the newly proposed HMSA with two multi-scale methods in Table 3: CTMDI [60] and ViT-CoMer [61]. The results indicate that the proposed HMSA surpasses other multi-scale approaches in all evaluation metrics.

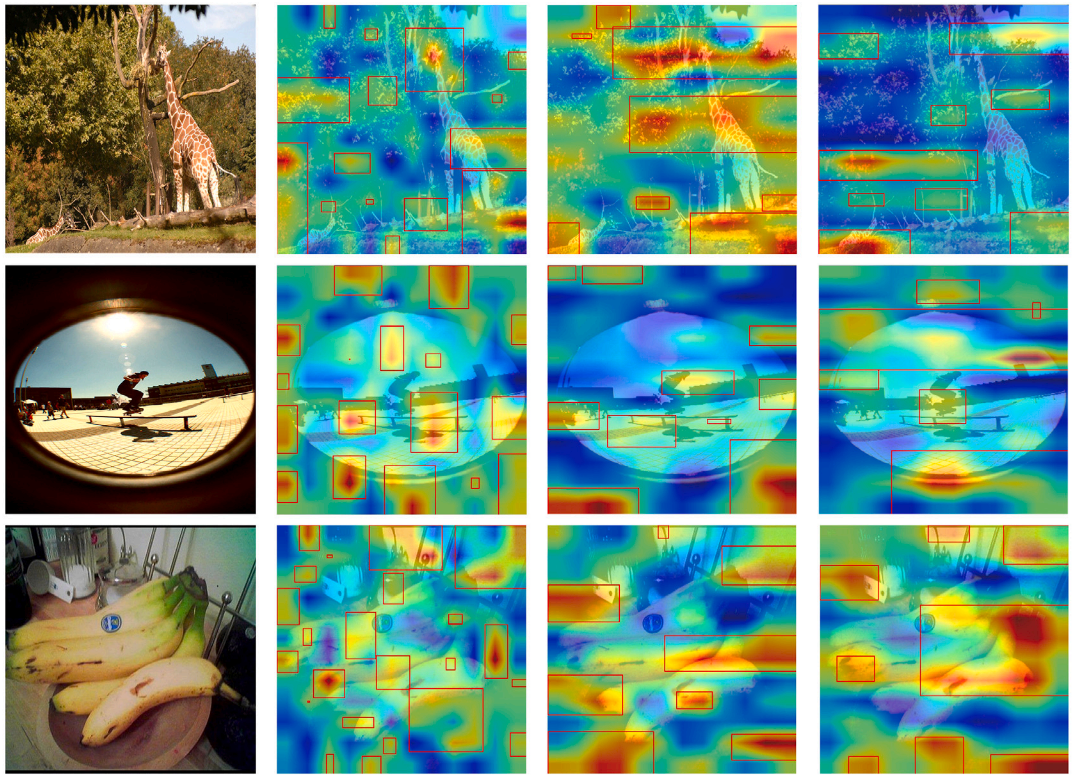

Fig. 4. Comparison results of CCMID with Co-attention and Merged attention two intermodal interaction modalities.

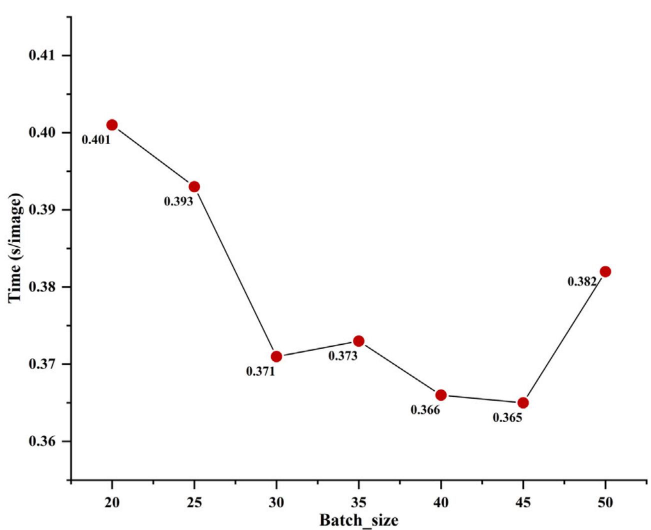

Fig. 5. Training time (in seconds) for each image under different Batch-size.

For instance, HMSA improves the BLEU4 score by 1.0% and 0.2%, respectively, and improves the CIDEr score by a margin of 3% and 1%. These results demonstrate that HMSA is more effective in capturing the details and contextual relationships of visual features, leading to more accurate captions. Fig. 4 presents a hierarchical visualization of the HMSA during the image extraction process, where the red boxes highlight areas with higher attention, further validating the model's strategy of focusing on local information at shallow layers and global information at deeper layers.

Table 3
Comparison results of CTMDI, ViT-CoMer and HMSA.

Methods	B-4	M	R-L	C	S
CTMDI [60]	37.1	27.5	56.8	117.4	20.4
ViT-CoMer [61]	37.9	28.0	57.0	119.4	21.3
Ours	**38.1**	**28.2**	**57.2**	**120.4**	**21.5**

Table 4
Comparison results of CTMDI, ViT-CoMer and HMSA.

Methods	B-4	M	R-L	C	S	Param/M
Co-attention [48]	38.1	28.3	57.8	118.4	20.7	154.34
Merged attention [49]	38.9	28.5	58.0	121.5	21.8	151.96
CCMID attention	**39.4**	**28.7**	**58.1**	**122.3**	**22.3**	**151.60**

4.3.3. CCMID module analysis

Table 4 presents a comparative analysis of the three methods in the captioning process. The numerical indicate that our approach is superior to other cross-modal fusion methods across all evaluation metrics. For the BLEU4 metric, our method improves by 1.3% compared to the collaborative method and by 0.5% compared to the direct merging method. In terms of the CIDEr metric, our method outperforms the collaborative method by 3.9% and the direct merging method by 0.8%. Additionally, CCMID has the fewest training parameters among the three methods.

4.4. Compare with SOTA models

In recent years, image captioning technology has made significant progress, with various new models continuously setting performance records. We compared the proposed model with some recent state-of-the-art (SOTA) models. Anderson et al. [30] introduced the Up-Down

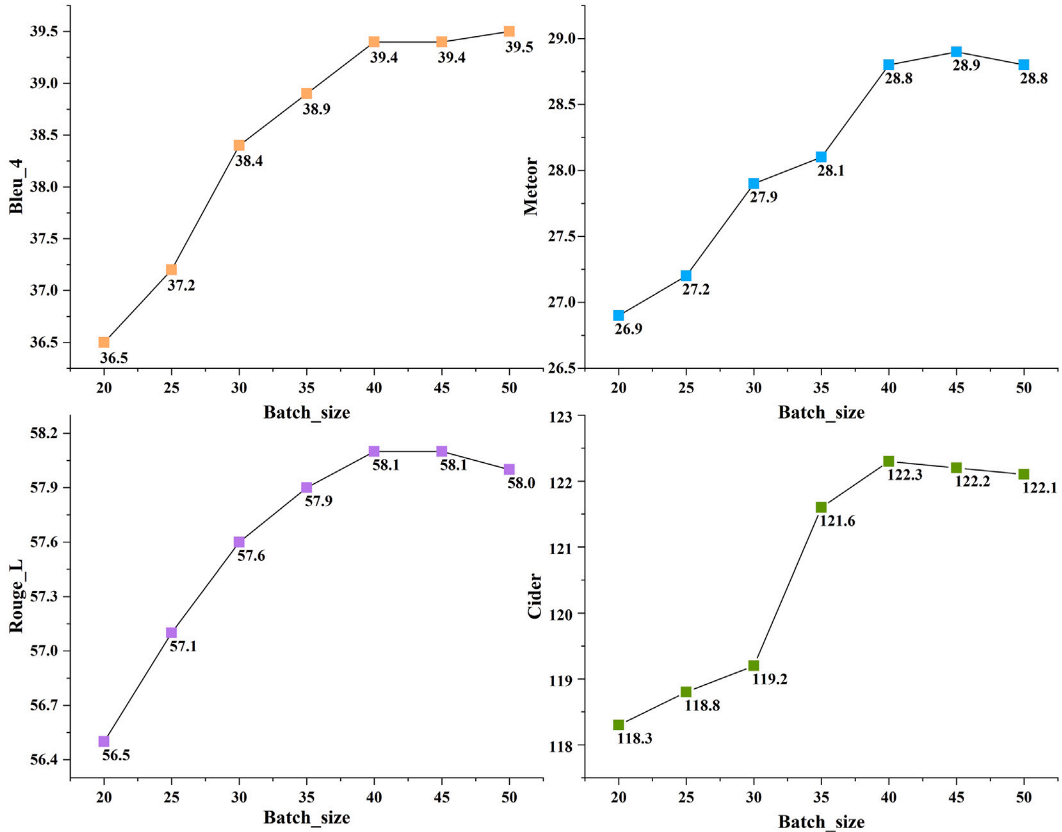

Fig. 6. Evaluation of caption quality generated by models under different Batch-size.

Table 5
Comparison with the SOTA model. Results are organized by the publication date of the papers, and all models were trained using the cross-entropy loss function.

Models	B-4	M	R-L	C	S
Up-Down [30]	36.2	27.0	56.4	113.5	20.3
AoANet [62]	37.2	28.4	57.5	119.8	21.3
X-Transfomer [63]	37.0	28.7	57.5	120.0	21.8
DSPLTN [64]	37.6	28.7	57.5	119.7	21.5
P+D attention [65]	37.3	28.6	57.6	117.3	–
GAT [41]	37.8	28.5	57.6	119.8	21.8
PureT [17]	37.4	28.9	57.7	121.6	22.0
SCD-Net [25]	37.3	28.1	58.0	118.0	21.6
CA-Captioner [31]	38.3	**28.8**	57.7	122.0	21.8
ACC [66]	37.6	28.6	57.6	122.0	21.8
CM-SC [67]	37.4	**28.8**	57.9	120.3	21.8
ViPCap [68]	37.7	28.6	–	**122.9**	21.9
Ours	**39.4**	28.7	**58.1**	**122.3**	**22.3**

model, which includes a visual attention mechanism and demonstrated significant performance in the image captioning task. Huang et al. [62] developed AoANet, which further enhanced feature capturing capabilities through the self-attention mechanism. X-Transformer [63] introduced a framework based on multi-level feature fusion, while Jiang et al. [64] proposed the DSPLTN model, which adopted a pyramid structure and combined multi-scale feature maps for image-based captioning. In the research by Wang et al. [17], the PureT model integrated a Transformer architecture, optimizing both feature extraction and information interaction. CA-Captioner [31] introduced a fusion mechanism that incorporated positional encoding, effectively enhancing the representation of local features. Attribute Controlled fashion image captioning (ACC) [66] introduces semantic attributes and utilizes multimodal image information to achieve style control in subtitle generation, demonstrating excellent performance in both general and fashion images. CM-SC [67] attention mechanism leverages a flexible cross-variance variant to efficiently model high-order interactions between visual and textual modalities, and can be seamlessly integrated into LSTM or Transformer frameworks, significantly enhancing caption quality while maintaining computational efficiency. ViPCap [68] is a lightweight image captioning model that maps retrieved text into visual prompts, effectively enhancing visual understanding while using fewer parameters than most existing lightweight models.

The results in Table 5 indicate that our model attained the highest scores for both BLEU4 score and CIDEr score, reaching 39.4% and 122.3%, respectively, surpassing several previous SOTA models. Specifically, our model improved the BLEU4 metric by 1.0% compared to PureT and by 0.8% compared to SCD-Net. Moreover, the model also performed well on the ROUGE-L and SPICE metrics, further proving the effectiveness of our proposed multimodal interaction mechanism in enabling deep-level information exchange between images and text.

4.5. Inference speed evaluation

To further evaluate the training efficiency and scalability of our model, we conducted experiments under different batch sizes and input resolutions to analyze their impact on both training time and captioning performance. Larger batch sizes typically improve computational throughput, while higher input resolutions may lead to richer feature representations but also increase memory and time costs.

We measured subtitle quality and inference speed in batches of 20, 25, 30, 35, 40, 45, and 50. The experimental results in Figs. 5 and 6 indicate that increasing the batch size leads to a continuous improvement in subtitle performance until the batch size reaches 40, after which all indicators tend to stabilize. For example, when the batch size is 45 or 50, the improvement is less than 0.3% or decreases, which may be due to memory saturation and dynamic optimization reduction. In terms of efficiency, the average processing time for each image decreases as the batch size increases, reaching its minimum

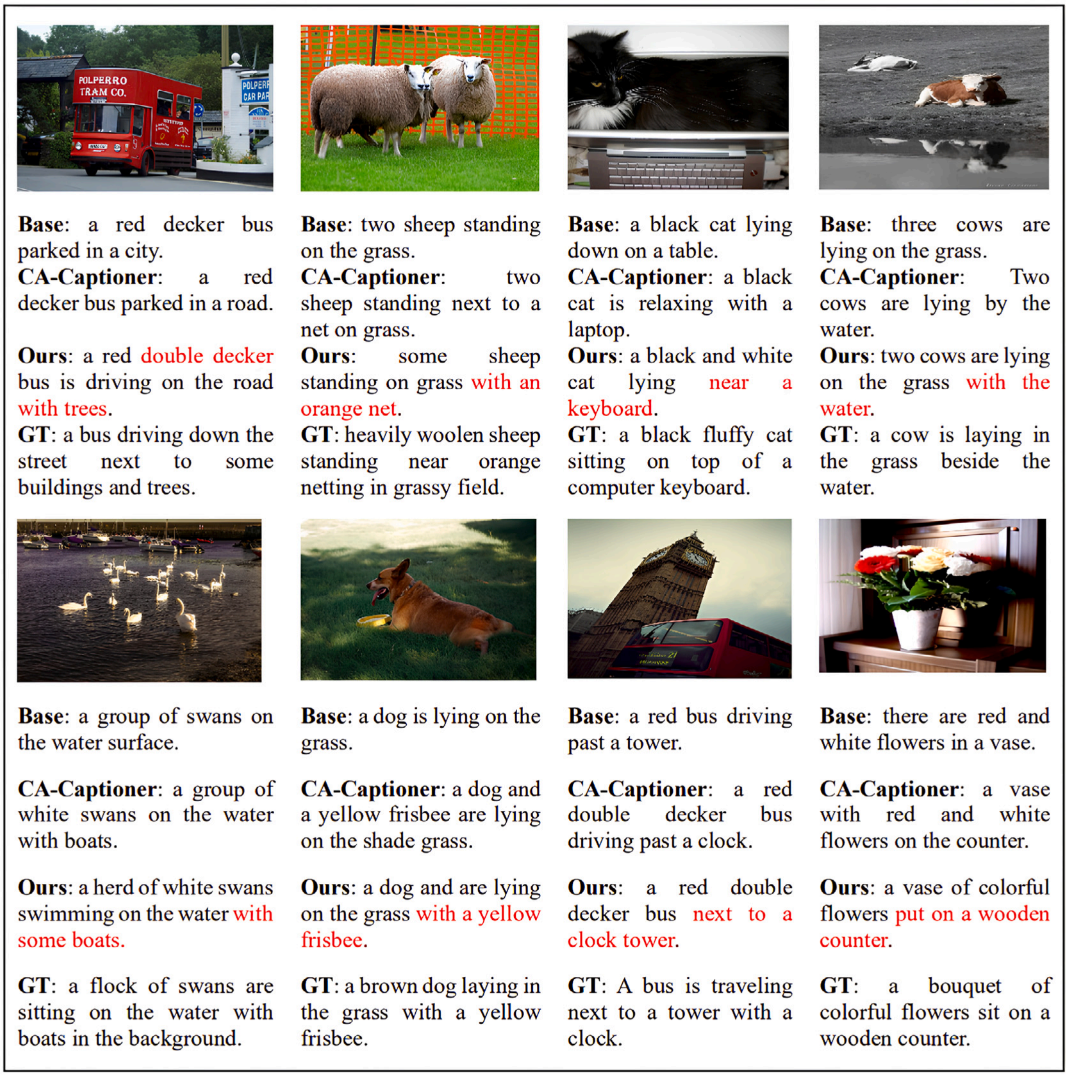

Fig. 7. Instances of sentences produced by the baseline model, the CA-Captioner model, the model proposed in this paper and ground-truths (GT).

value at a batch size of 40, followed by a slight increase in latency. Therefore, a batch size of 40 provides the best performance between speed and subtitle quality, making it a suitable choice to balance actual deployment requirements and high performance.

In the Table 6, we compared the training time and generation quality of the model under two input resolutions of 224[69] and 384. The 224 × 224 input resolution significantly reduces the time per image (by over 56.8%), making it more efficient in terms of computation. However, this efficiency comes at the cost of substantially lower captioning performance across all evaluation metrics. In contrast, using 384 × 384 images may require more computation time, but it will result in significantly better performance in semantic relevance and fluency. This performance gap suggests that higher resolution inputs provide richer visual representations, which are especially important for dense scene understanding and fine-grained object relationships, both of which are critical in the image captioning task.

Table 6
Time and generation performance corresponding to resolutions of 224 and 384.

Input-resolution	Time/image	B-4	M	R-L	C
224 × 224	0.158 s	26.4	18.3	36.0	105.8
384 × 384	0.366 s	39.4	28.7	58.1	122.3

4.6. Generated captions for presentation and analysis

To illustrate the influence of our method on caption generation more clearly, we analyzed the sentences produced by the baseline model, the CA-Captioner model, and our proposed model. The figures reveal that our model effectively captures the relationships among objects and between objects and the overall scene. Fig. 7 presents a comparison of the captions generated by the baseline model, the CA-Captioner model, and our proposed HI-Captioner model, alongside the ground-truth instances. The results show that the HI-Captioner model generates more accurate captions. In contrast to the baseline model, our model focuses on the main subjects in the image while also highlighting the spatial relationships between various elements and elaborating on them in the caption.

For example, in the first image, our model captures the relationship between the “bus” and the background elements “buildings” and “trees”, identifying their spatial relationship as “next to”, while also providing detailed captions of the “bus” features. In the last image, the model accurately captures the spatial relationship between the “cow” and the environment, including “water” and “grass”. Compared to the captions generated by the baseline model, our captions exhibit better hierarchical structure and coherence.

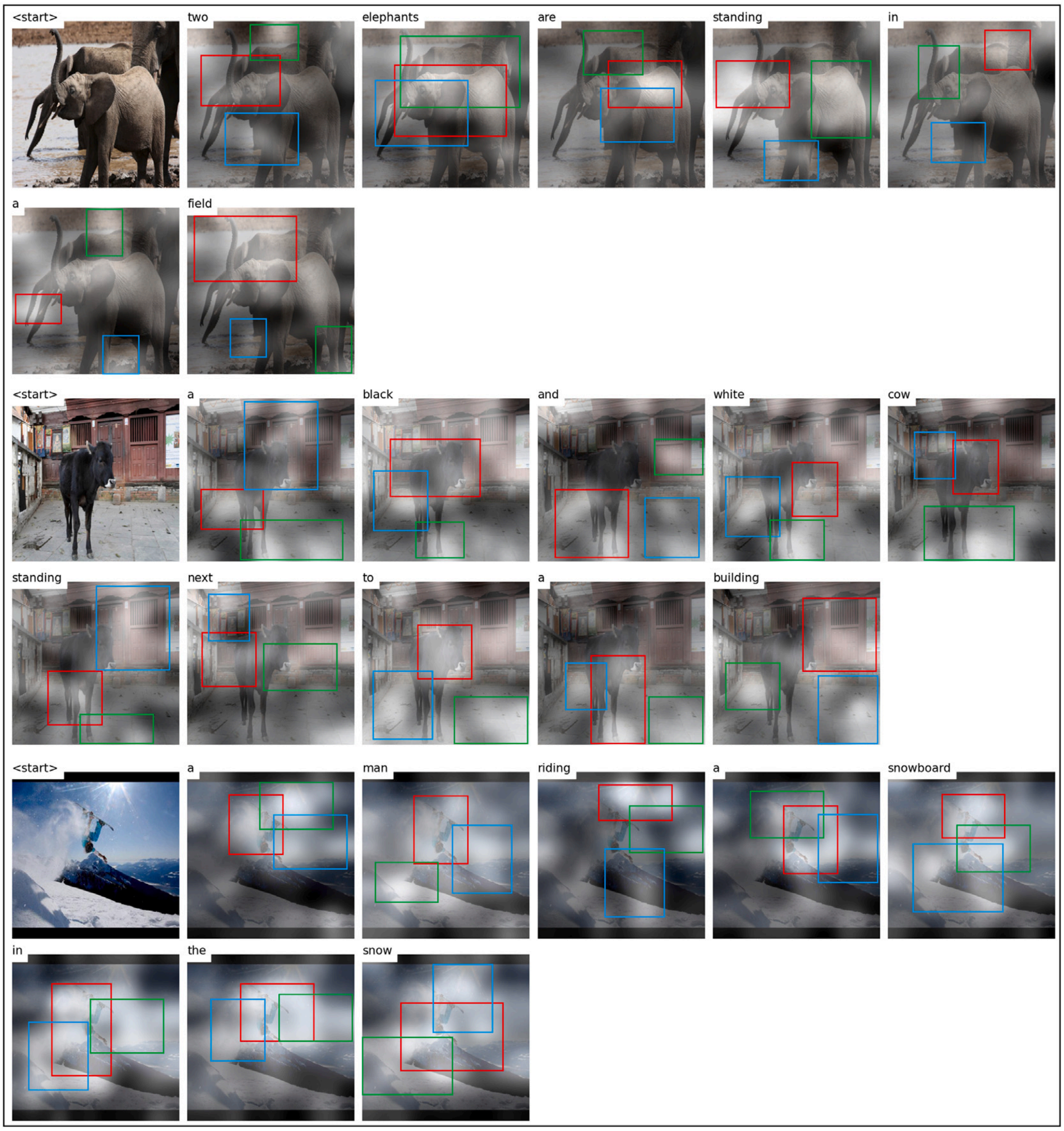

Fig. 8. HI-Captioner generates a caption visualization map, with red, green and blue being the top three regions of the attention score in that order.

Table 7
Flickr datasets utilized in the experiments along with their respective splits.

Dataset	Train	Val	Test	All
Flickr30k	28 000	1000	1000	30 000
Flickr8k	6000	1000	1000	8000

Table 8
Results of the Flickr dataset.

Dataset	Model	B-4	M	R-L	C	S
Flickr8k	Base	40.1	19.2	23.3	116.5	14.7
	Ours	**41.3**	**19.5**	**24.6**	**120.3**	**15.2**
Flickr30k	Base	71.6	40.7	44.3	121.9	19.0
	Ours	**73.8**	**44.5**	**47.6**	**126.7**	**21.5**

4.7. Robustness experiment

In addition to the MSCOCO dataset, we also selected Flickr8k [70] and Flickr30k [71] as validation datasets. The Flickr datasets are moderately sized and have been widely used in early image captioning research. Their splits are shown in Table 7. As presented in Table 8, our proposed HI-Captioner achieved favorable results on both Flickr datasets, with the BLEU4 score increasing by 1.2% and 2.2%, and the CIDEr metric increasing by 3.8% and 4.8%, respectively. Table scores highlight the impressive generalizability and robustness of our proposed model.

As analyzed in the feature activation map in Fig. 3, through the HSAM module, the model can simultaneously process image features from different scales, enhancing its ability to capture image details. This is particularly effective during the caption generation process, where the model can center on important regions of the image. The heatmaps in Fig. 8 represent the model's attention focus, with the areas receiving higher attention highlighted by different colored boxes. These images demonstrate that, as the text is generated, the model's attention shifts to the corresponding parts of the image, resulting in more accurate text captions.

5. Conclusion

This paper introduces a new image captioning model called HI-Captioner, which is founded on multi-level information fusion and cross-modal interactive decoding. By introducing the Hierarchical Multi-Scale Attention (HMSA) mechanism and Hierarchical Positional Encoding (HPE), the model can flexibly adjust attention weights at different levels, and better capture the multi-level features and positional information of images in a hierarchical manner. Additionally, the proposed Cascading Cross-Modal Interactive Decoding (CCMID) combines self-attention and cross-attention to achieve deep information interaction between the visual and language modalities. The experimental findings demonstrate that the proposed model attains leading performance on several evaluation metrics, particularly excelling on the MSCOCO dataset with BLEU4 and CIDEr scores of 39.4% and 122.3%, respectively, surpassing existing SOTA models.

6. Discussion

Through a series of ablation studies and visualization analyses, our proposed method has demonstrated significant advantages in capturing fine-grained image details and generating coherent, natural language captions. It is particularly effective in accurately describing salient content within complex visual scenes. Moreover, the model exhibits enhanced generalization ability as well as notable robustness and flexibility in cross-modal information interaction. Despite these improvements, the extraction and utilization of multi-scale image features in end-to-end image captioning remain areas that warrant further investigation. In future work, integrating HI-Captioner with large-scale pre-trained multimodal language models (e.g., BLIP-2, Flamingo, GPT-4V) presents a promising direction for improving caption quality. These models possess powerful cross-modal reasoning capabilities that could complement HI-Captioner's structured attention and encoding pipeline. Potential integration strategies include leveraging these models as external semantic encoders, applying prompt-based fine-tuning techniques, or incorporating them into a hybrid encoder–decoder architecture. Additionally, their ability to understand open-domain knowledge and produce diverse linguistic expressions may help address existing limitations in generating more human-like and contextually appropriate descriptions [72–74].

CRediT authorship contribution statement

Wenjing Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Xue Li:** Validation, Supervision, Resources, Project administration, Methodology, Conceptualization. **Ziyang Li:** Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jiong Yu:** Supervision, Software, Resources, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Pengcheng Chen:** Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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